Introduction to Exploratory Data Analysis

Course 4. Outlier detection

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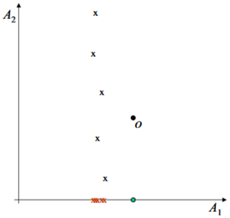
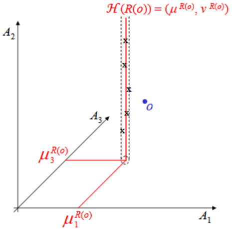
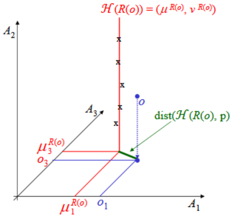
Subspace outlier detection



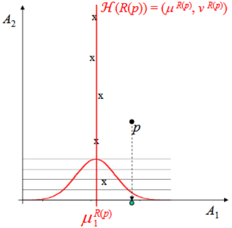
(a) The general idea of finding outliers in subspaces.



(b) Illustration of the subspace hyperplane of a reference set $R(o)$.



(c) Illustration of the distance between a point o and a subspace hyperplane $\mathcal{H}(R(o))$.

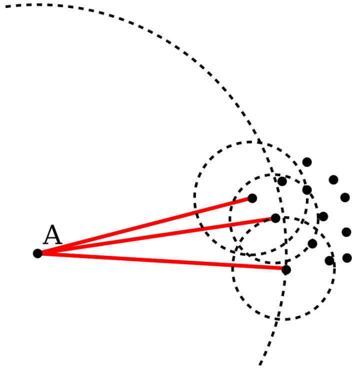


(d) Gaussian distribution of the distances of all points in $R(p)$ to $\mathcal{H}(R(p))$.

Local Outlier factor

Unsupervised method for multidimensional data.

Compare the density of the region a sample $\boldsymbol{p} \in \boldsymbol{X}$ lives in with the density of his nearest k neighbors regions.



Local Outlier Factor

Call ***k-distance*** of a sample $\boldsymbol{p}$ the distance $d(\boldsymbol{p}, \boldsymbol{o})$ between $\boldsymbol{p}$ and another signal $\boldsymbol{o}$ such that

1. for at least k samples $\boldsymbol{o}' \in \boldsymbol{X} \setminus \{\boldsymbol{p}\}$, $d(\boldsymbol{p}, \boldsymbol{o}') \leq d(\boldsymbol{p}, \boldsymbol{o})$ and
2. for at most $k-1$ samples $\boldsymbol{o}' \in \boldsymbol{X} \setminus \{\boldsymbol{p}\}$, $d(\boldsymbol{p}, \boldsymbol{o}') < d(\boldsymbol{p}, \boldsymbol{o})$.

Define the ***k-distance neighborhood*** of $\boldsymbol{p}$ the set of samples found at a distance smaller than *k-distance* from $\boldsymbol{p}$

$$N_k(\boldsymbol{p}) = \{\boldsymbol{q} \in \boldsymbol{X} \setminus \{\boldsymbol{p}\} \mid d(\boldsymbol{p}, \boldsymbol{q}) \leq \text{k-distance}(\boldsymbol{p})\} \tag{1}$$

Local Outlier Factor

Define the *reachability distance* from $\boldsymbol{p}$ to $\boldsymbol{o}$

$$\text{reach-dist}_k(\boldsymbol{p}, \boldsymbol{o}) = max(\text{k-distance}(\boldsymbol{o}), d(\boldsymbol{p}, \boldsymbol{o})) \tag{2}$$

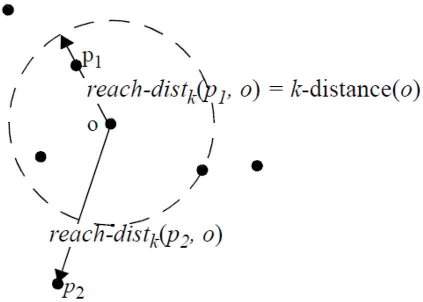


Image source: M. Breunig, H. Kriegel, R. Ng, J. Sander, LOF: Identifying Density-Based Local Outliers, SIGMOD, 2000

Local Outlier Factor

Define the *local reachability density* of $\boldsymbol{p}$

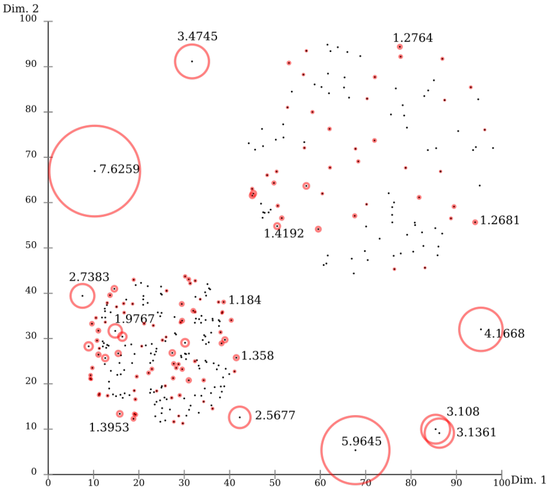
$$lrd_k(\boldsymbol{p}) = \left(\frac{\sum_{\boldsymbol{o} \in N_k(\boldsymbol{p})} \text{reach-dist}_k(\boldsymbol{p}, \boldsymbol{o})}{|N_k(\boldsymbol{p})|} \right)^{-1} \tag{3}$$

Define the LOF score of $\boldsymbol{p}$

$$LOF_k(\boldsymbol{p}) = \frac{\sum_{\boldsymbol{o} \in N_k(\boldsymbol{p})} \frac{lrd_k(\boldsymbol{o})}{lrd_k(\boldsymbol{p})}}{|N_k(\boldsymbol{p})|} \tag{4}$$

If a sample has small local density of his neighbors have large density, the score will be close to 1 and the signal is considered an anomaly.

Local Outlier factor



Multidimensional signals

Low dimensional spaces: All dimensions are relevant and distances (such as Euclidian distance) can give clues on a sample's normality.
High dimensional spaces: Data points are equidistant. For each data point, the distance between it and the nearest data point is similar to the distance to the furthest point.

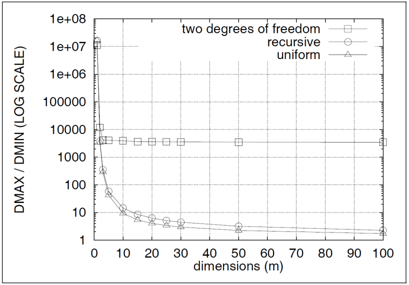


Fig. 7. Correlated distributions, one million tuples.

Image source: Beyer, K., Goldstein, J., Ramakrishnan, R., Shaft, U. When Is "Nearest Neighbor" Meaningful? Database Theory — ICDT'99. Lecture Notes in Computer Science, vol 1540. Springer

Angle-based Outlier Detection (ABOD)

Because anomalies lie in the margin of the data space, the angle between an anomaly and any other two samples will be less than the angle between a normal signal and any other two normal samples.

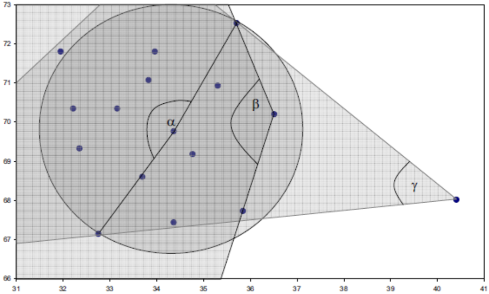


Image source: H. Kriegel, M. Schubert, A. Zimek, Angle-based outlier detection in high-dimensional data

Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining, 2008

ABOD

Let $\boldsymbol{A} \in \boldsymbol{X}$ be a sample. ABOD computes the angles between $\boldsymbol{A}$ and all pairs of points $(\boldsymbol{B}, \boldsymbol{C})$, where $\boldsymbol{B} \in \boldsymbol{X} \setminus \{\boldsymbol{A}\}$ și $\boldsymbol{C} \in \boldsymbol{X} \setminus \{\boldsymbol{A}, \boldsymbol{B}\}$.

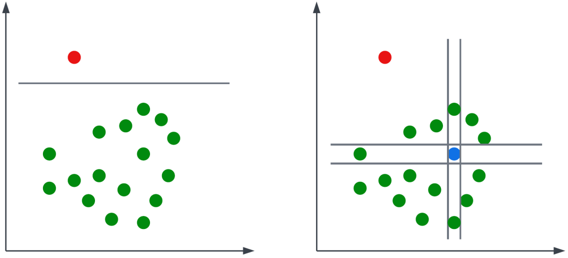
The less the distance from $\boldsymbol{A}$ to the two points, the more the angle will vary more due to noise. So, normalize the angle with the product of the squared distances from $\boldsymbol{A}$ to the two points.

The anomaly score is computed as the variance of the angles between $\boldsymbol{A}$ to all pairs of points in $\boldsymbol{X}$,

$$ABOF = VAR_{\boldsymbol{B},\boldsymbol{C} \in \boldsymbol{X}} \left(\frac{\langle \overline{\mathrm{AB}}, \overline{\mathrm{AC}} \rangle}{\| \langle \overline{\mathrm{AB}} \rangle \|^2 \cdot \| \langle \overline{\mathrm{AC}} \rangle \|^2} \right) \tag{5}$$

Isolation forest

Unsupervised method for multidimensional data.



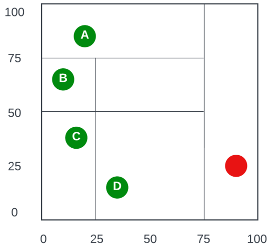
Isolating an anomaly — Isolating a normal sample

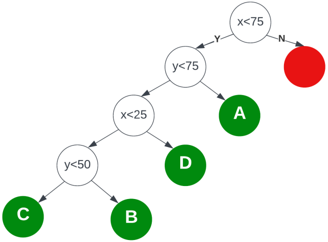
Isolation is done using one dimension at a time.
Neither the relevant dimensions nor the optimal threshold are known in advanced (i.e. where to place the separation line/hyperplane).

Isolation forest

Mechanism: Randomly pick a dimension q and a threshold p (in the [min, max] interval of that dimension) and recursively construct a decision tree.

- Internal nodes of the tree represent the tests $q < p$;
- Leaves represent the data samples.





Isolation forest

- In the decision iTrees, anomalies are found at lower heights that the normal signals (closer to the root of the tree).
- A single tree is not sufficient for proper anomaly separation $\implies$ *Forest* (≈ 100 iTrees).
- For the anomalies to be more obvious, a small number of samples is better $\implies$ *subsampling* (256 samples for each iTree, randomly chosen).

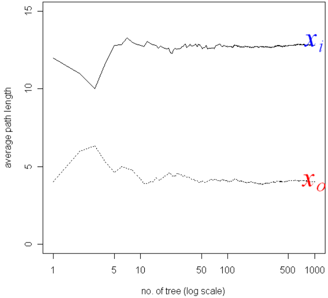


Image source: Liu, Fei Tony, Ting, Kai Ming, Zhou, Zhi-Hua. Ïsolation Forest". 2008 Eighth IEEE International Conference on Data Mining

Isolation forest

Training - Build the iForest.

Anomaly score - Based on the the mean height value in all iForest trees. Needs to be normalized w.r.t. mean leaf nodes height.

Mean leaf nodes height is equivalent to the average length of unsuccessful search in a binary tree of n nodes:

$$c(n) = 2H(n-1) - 2(n-1)/n \tag{6}$$

$H(i)$ is the Harmonic number, $H(i) = ln(i) + \gamma$, where γ is Euler's constant. The anomaly score for sample $\boldsymbol{x}$ is

$$s(\boldsymbol{x}, n) = 2^{-\frac{E(h(\boldsymbol{x}))}{c(n)}} \tag{7}$$

If $s(\boldsymbol{x}, n)$ is close to 0, $\boldsymbol{x}$ is normal, if it is close to 1, $\boldsymbol{x}$ is an anomaly.

Source: Liu, Fei Tony, Ting, Kai Ming, Zhou, Zhi-Hua. Ïsolation Forest". 2008 Eighth IEEE International Conference on Data Mining

Isolation forest

Testing
So far, we only created the iForest. No classification yet.

Each iTree with his own set of random dimensions and thresholds; each iTree with its different (randomly chosen) set of samples.

To compute the anomaly scores, each signal is passed through each tree in the iForest until he reaches a leaf. The height of those leaves are averaged and the (7) is used to compute the score.

1. Anomalies are responsible for most of the variance in the data.
2. Anomalies lie on the margins of data cloud.
3. Anomalies are found in lower-density regions of the data.
4. Anomalies manifest in some sub-space of the data.
5. Anomalies are easier to separate from the rest of the data than normal samples are.
6. If data is modeled using statistical tools, the model will better represent normal samples and will do worse with outliers.